DISCUSSION

White button mushrooms are small to medium in size with caps averaging 2-7 centimeters in diameter and are attached to short, truncated stems. The smooth white caps are rounded, firm, and spongy. When bruised, the white flesh will turn light pink and will then transform into brown. Underneath the unopened caps, there are many small, light brown gills that are hidden with a white veil and these gills produce dark brown spores. The short white stem is also edible, dense, thick, and smooth. When raw, White button mushrooms are mild with a crisp texture, and when cooked, they develop an earthy flavor with a tender, chewy texture.

The FASTA sequence of *Agaricus bisporus* was retrieved from NCBI (Accession No ACC18877) along with the translated protein sequence. The BLAST program was run, Multiple sequence alignment was done, a Phylogenetic tree was constructed, and SWISS-MODEL values were observed. All these parameters suggested that was the closest template. This template has a sequence homology percentage of 48.87%. Three-dimensional structure was predicted by an online 3D structure predicting server, SWISS-MODEL. The predicted structure was validated by ERRAT, Ramachandran plot, and VERIFY 3D values. These findings showed that the overall ERRAT quality factor output in the minimized AbL model was 86.558%. ERRAT showed the percentage of non-bonded interactions between different atoms for which the estimated error score fell below the 90% rejection threshold.

Some residues are present in the rejection area because they are not correctly crystallized in the template model during X-Ray Crystallography, and they are also causing disturbance here. Ramachandran plot was obtained from the PROCHECK for the evaluation of backbone conformations, and stereochemical quality of the modeled structure by Φ-ψ plot, Ramachandran plot for AbL (Fig. 6) revealed that 85.9% with 359 residues are present in the most favored regions, 12.4% with 52 residues in the additional allowed regions, and 1.2% residues are present in the generously allowed and the disallowed areas. Total residues are 503, no. of non-glycine and non-proline residues are 418, no. of end residues (excl. Glycine and Proline) are 2, no. of glycine residues are 36 and no. of proline residues are 44. A good quality structure has more than 90% residues in the most favored regions. VERIFY 3D results showed that 83.20% of the structure residues have averaged 3D-1D score >= 0.1. At least 80% of the amino acids have scored >=0.1 in the 3D/1D profile. Obtained structure was aligned with the template, and a 1.38 Å RMSD value was obtained, which showed that the structure predicted is much similar and was built according to the template used.

AbL has a molecular weight of 50.064 kDa, and it contains a total of 520 amino acids having 2 domains that are connected via a double-stranded linker peptide. Each domain has a 20 Å radius and comprises 3 four-stranded β sheets and six helices. 26% of residues are found in β sheets, while 33% are found in helices. The antiparallel strand of one β sheet is constructed by the N and C termini, which are situated in domain I. The sequence of C-terminal part (residues 1–17 and 232-419) makes up the majority of domain I, whereas the N-terminal region makes up the majority of domain II (residues 22–227) (Schönbrunn *et al.*, 1996).

The process of virtual screening was done with the help of PyRx. This includes the selection of ligand. The ligand was selected from protein data bank PDB. All the ligands from that were selectedThe screening results were observed, and the ligands with low binding free energies and RMSD values was selected. Ligand selected include the PubChem ID: GLU 753. It has RMSD value less than 2 Å. From 1.412 million chemicals in three databases (i.e., FDA-approved drugs, Sigma database, and ChemBridge database), putative Glycerol was found using a molecular docking-based virtual screening technique.

We have to obtain the Agaricus bisporus laccase 3D structure from the Protein Data Bank (PDB) or, in the absence of one, use homology modelling to predict the structure and remove any ions, water molecules, or other tiny molecules that aren't a part of the enzyme to prepare the protein structure. Assign atom kinds and include any extra atoms required for a full structure. Build the 3D structure of glycerol using molecular modelling tools or obtain it from chemical databases. Glycerol should be converted to the proper 3D structure format, such as PDB, SDF, or MOL2, and its shape should be optimised by reducing energy.

Select a reliable molecular docking programme for protein-ligand docking, such as AutoDock, AutoDock Vina, Glide, GOLD, or another popular programme. It is important to determine the location on Agaricus bisporus laccase's active site where glycerol is thought to bind. This can be accomplished using the catalytic location of the enzyme or any other structural data that is available. We have to create the docking parameters, which should include the search space (the area where Glycerol can attach), the number of docking poses to be produced, and any special requirements or preferences for the docking algorithm. Run the

molecular docking simulation to determine the most advantageous Glycerol binding pose(s) within the enzyme's active site.

With each docking pose, the docking software produces a docking score that corresponds to the projected binding affinity. We examined the docking findings to determine the best docking pose(s) based on the interactions created between Glycerol and Agaricus bisporus laccase and the docking scores. We utilised molecular visualisation software to see the binding positions and examine the interactions between the enzyme and glycerol, such as hydrogen bonds and hydrophobic interactions. It was done with great care when interpreting the data because molecular docking predictions aren't necessarily accurate depictions of actual binding interactions. Different methods were used like X-ray crystallography or other binding assays to do experimental validation to verify the expected binding mechanism and affinity between Glycerol and Agaricus bisporus laccase.

. In Agaricus bisporus, laccase predominantly contributes to the lignin breakdown and melanin synthesis processes, which are crucial for the mushroom's growth and development. Agaricus bisporus is a saprophytic fungus that breaks down complex organic matter to gain nutrients, however, it predominantly uses decaying plant matter and pollution degradation. Laccase plays an important role in the degradation of certain environmental pollutants particularly aromatic compounds. Effluents from numerous industrial activities often contain hazardous and resistant organic contaminants. Laccases can be utilised in bioremediation techniques to break down complex organic molecules into simpler and less hazardous chemicals, which can be used to cleanse toxic industrial effluents. Laccase can be used to treat organic waste, including municipal solid waste, agricultural waste, and waste from food production. Laccase can help lessen the negative effects of such waste on the environment by decomposing organic compounds.